

GLASGOW, MT, USA

WMO: 727680

Lat: 48.2067N

Lon: 106.6256W

Elev: 2278

StdP: 13.53

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94008

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-14.6	-25.6	1.4	-19.3	-19.7	2.0	-13.5	32.2	22.7	28.3	16.4	7.1	310	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.0	94.6	64.9	91.2	64.1	87.5	63.0	68.9	85.4	67.0	83.8	65.3	82.0	11.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	96.0	75.0	61.3	88.2	72.5	59.3	82.1	71.2	34.4	84.9	32.7	83.6	31.2	81.7	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.5	25.3	22.4	DB	-27.2	100.9	6.5	2.5	-31.9	102.7	-35.7	104.2	-39.4	105.7	-44.1	107.5	(4)
(5)			WB	-26.2	72.7	5.3	2.8	-30.0	74.7	-33.1	76.3	-36.1	77.9	-40.0	79.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	14.2	17.4	30.3	44.0	54.5	63.7	72.0	70.4	59.5	44.8	29.8	17.3		
	DBStd	23.01	15.31	14.39	14.08	9.33	8.33	6.84	6.06	6.83	8.80	9.34	12.59	14.06		
	HDD50	4759	1110	913	615	219	49	2	0	0	20	209	608	1014		
	HDD65	8460	1575	1333	1075	629	338	103	11	28	207	626	1056	1479		
	CDD50	2317	0	0	4	40	188	413	681	633	307	49	2	0		
	CDD65	543	0	0	0	0	12	64	227	196	43	1	0	0		
	CDH74	7198	0	0	0	35	255	844	2873	2486	661	44	0	0		
	CDH80	3127	0	0	0	5	60	275	1348	1172	259	8	0	0		
	WSAvg	10.4	9.4	9.5	11.2	12.1	11.8	10.9	10.3	10.5	10.2	10.5	9.6	9.2		
	PrecAvg	11.3	0.4	0.3	0.4	0.8	1.7	2.4	1.9	1.3	0.9	0.6	0.4	0.4		
	PrecMax	17.8	1.3	1.0	1.3	2.0	3.7	5.4	5.9	5.7	4.1	3.1	1.5	1.0		
(15) Precipitation	PrecMin	6.7	0.0	0.0	0.0	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	2.8	0.3	0.2	0.2	0.5	0.9	1.3	1.5	1.2	0.9	0.6	0.3	0.2		
	0.4%	DB	48.0	52.7	68.4	79.0	86.1	92.9	99.4	99.0	92.9	80.1	64.3	49.0		
	MCWB	39.8	41.5	49.6	54.4	60.8	65.4	65.8	65.0	63.0	56.9	48.8	39.3			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2%	DB	41.6	44.5	60.1	71.7	80.6	86.9	95.3	94.4	87.5	71.1	56.7	42.6		
	MCWB	36.0	37.2	45.7	52.0	58.6	64.2	65.7	64.3	59.9	53.0	44.7	35.7			
	5%	DB	38.0	39.6	54.3	66.2	76.1	82.6	91.8	90.9	81.9	65.2	50.5	38.6		
	MCWB	33.8	34.5	42.6	48.9	56.5	62.4	65.2	63.3	59.2	50.3	41.2	33.8			
	10%	DB	34.4	35.5	48.8	61.3	71.2	78.8	87.8	87.1	76.3	59.9	45.9	35.2		
	MCWB	31.3	32.3	39.7	46.6	53.9	60.8	64.2	62.4	57.1	47.8	38.7	31.8			
	0.4%	WB	39.9	42.4	50.2	55.8	63.2	70.7	72.8	69.7	66.4	58.4	50.2	39.9		
	MCDB	46.6	51.5	65.9	74.8	79.0	86.1	89.7	88.8	83.4	75.0	62.2	47.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.3	37.6	46.6	52.7	60.1	66.5	69.8	67.3	62.8	54.3	45.4	36.5		
	MCDB	41.1	43.6	59.3	69.5	75.9	81.6	85.7	85.4	79.8	68.7	56.3	42.0			
	5%	WB	34.0	34.8	43.3	50.2	57.9	64.3	68.0	65.6	60.5	51.1	41.8	34.1		
	MCDB	37.7	39.1	53.3	64.3	72.5	78.6	84.3	84.0	77.6	63.2	49.2	38.2			
	10%	WB	31.4	32.4	40.1	47.5	55.6	62.3	66.2	64.0	58.3	48.7	39.1	31.8		
	MCDB	34.6	35.7	48.0	59.6	68.6	75.7	82.5	82.3	75.0	58.7	45.5	35.3			
	5% DB	MDBR	17.5	18.2	20.2	23.7	23.9	23.3	27.0	27.4	25.6	22.3	19.5	17.3		
	MCDBR	19.8	22.0	29.0	33.0	31.4	29.8	32.6	32.6	33.9	31.0	26.6	20.8			
(36) Mean Daily Temperature Range	MCWBR	15.1	15.5	16.6	16.8	14.3	12.6	11.1	11.3	13.9	16.0	16.5	15.5			
	5% WB	MCDDBR	19.6	20.9	28.2	30.5	28.1	26.4	28.3	28.5	30.1	29.0	25.5	20.0		
	MCWBR	15.3	15.1	16.8	15.9	13.4	12.5	11.4	10.7	13.0	15.4	16.3	15.2			
(40) Clear-Sky Solar Irradiance	taub	0.250	0.256	0.272	0.318	0.341	0.350	0.360	0.368	0.322	0.283	0.258	0.241			
	taud	2.498	2.540	2.577	2.439	2.419	2.445	2.430	2.380	2.478	2.585	2.558	2.520			
	Ebn at Noon	272	295	305	297	292	289	284	277	282	280	267	263			
	Edh at Noon	19	23	26	33	35	35	35	35	29	22	18	16			
(44) All-Sky Solar Radiation	RadAvg	442	748	1148	1569	1889	2082	2225	1848	1338	854	507	341			
	RadStd	35	53	80	75	142	128	108	122	106	73	53	25			

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page